

$$v_c'(t) + \frac{1}{\tau} v_c(t) = \frac{1}{\tau T} \quad R = \frac{\tau}{T}$$

$$= e^{-\frac{t}{\tau}} \left(\int e^{\frac{t}{\tau}} \cdot \frac{1}{\tau T} dt + C \right)$$

$$= e^{-\frac{t}{\tau}} \left(\frac{1}{\tau T} \int e^{\frac{t}{\tau}} dt + C \right)$$

$$= e^{-\frac{t}{\tau}} \left(\frac{1}{T} e^{\frac{t}{\tau}} + C \right)$$

$$\frac{1}{T} + e^{-\frac{t}{\tau}} \cdot C$$

$$0 = \frac{1}{T} + C \quad C = -\frac{1}{T}$$

$$v_c(t) = \frac{1}{T} \left(1 - e^{-\frac{t}{\tau}} \right), \quad 0 \leq t < T$$

$$v_c'(t) + \frac{1}{\tau} v_c(t) = 0$$

$$v_c(t) = C e^{-\frac{t}{\tau}} \quad v_c(T) =$$

$$\begin{aligned}
 1) \quad \tau_1 &= R_1 C_1 = 10 \cdot 10^3 \Omega \cdot 1 \cdot 10^{-6} \text{ F} \\
 &= 10 \cdot 10^{-3} \text{ s} = 1 \cdot 10^{-2} \text{ s} \\
 \tau_2 &= R_2 C_2 = 4.7 \cdot 10^3 \Omega \cdot 1 \cdot 10^{-6} \text{ F} \\
 &= 4.7 \cdot 10^{-3} \text{ s} = 4.7 \cdot 10^{-3} \text{ s} \\
 G &= 1 + \frac{4.7 \text{ k}\Omega}{1 \text{ k}\Omega} = 1 + 4.7 = 5.7
 \end{aligned}$$

$$2) \quad \dot{v}_{c2}(t) = \frac{G v_{c1}(t)}{\tau_2} - \frac{v_{c2}(t)}{\tau_2} \quad \left| \quad v_{c1}(t) = \frac{\tau_2}{\tau_1} v_{c2}'(t) + G v_{c2}(t) \right.$$

$$v_{c2}''(t) = \frac{G}{\tau_2} v_{c1}'(t) - \frac{1}{\tau_2} v_{c2}'(t) = \frac{G}{\tau_2} \cdot \frac{1}{\tau_1} (v_{in} - v_{c1}(t)) - \frac{1}{\tau_2} v_{c2}'(t)$$

$$v_{c2}''(t) + \frac{1}{\tau_2} v_{c2}'(t) = \frac{G}{\tau_2 \tau_1} v_{in} - \frac{G}{\tau_2 \tau_1} v_{c2}(t)$$

$$v_{c2}''(t) + \frac{1}{\tau_2} v_{c2}'(t) = \frac{G}{\tau_2 \tau_1} v_{in} - \frac{G}{\tau_2 \tau_1} \left(\frac{\tau_2}{G} v_{c2}'(t) + G v_{c2}(t) \right)$$

$$v_{c2}''(t) + \frac{1}{\tau_2} v_{c2}'(t) = \frac{G}{\tau_2 \tau_1} v_{in} - \frac{1}{\tau_1} v_{c2}'(t) + \frac{1}{\tau_2 \tau_1} v_{c2}(t)$$

$$v_{c2}''(t) + \left(\frac{1}{\tau_2} + \frac{1}{\tau_1} \right) v_{c2}'(t) - \frac{1}{\tau_2 \tau_1} v_{c2}(t) = \frac{G}{\tau_2 \tau_1} v_{in}$$

$$3) \quad v_{c2}(t) = \frac{1}{\tau_2} e^{-\frac{t}{\tau_2}} u(t) \quad (\text{from section 4.1, } \tau = \tau_1)$$

$$v_{c2}'(t) + \frac{1}{\tau_2} v_{c2}(t) = \frac{G}{\tau_2 \tau_1} e^{-\frac{t}{\tau_1}} \quad t \geq 0$$

$$I(t) = \frac{t}{\tau_2}$$

$$v_{c2}(t) = e^{-\frac{t}{\tau_2}} \left(\int e^{\frac{t}{\tau_2}} \cdot \frac{G}{\tau_2 \tau_1} e^{-\frac{t}{\tau_1}} dt + C \right)$$

$$= e^{-\frac{t}{\tau_2}} \left(\frac{G}{\tau_2 \tau_1} \int e^{\frac{t}{\tau_2} - \frac{t}{\tau_1}} dt + C \right)$$

$$= e^{-\frac{t}{\tau_2}} \left(\frac{G}{\tau_2 \tau_1} \int e^{\frac{t(\tau_1 - \tau_2)}{\tau_1 \tau_2}} dt + C \right)$$

$$\frac{t}{\tau_2} - \frac{t}{\tau_1} = \frac{\tau_1 t - \tau_2 t}{\tau_2 \tau_1} = \frac{t(\tau_1 - \tau_2)}{\tau_2 \tau_1}$$

$$= e^{-\frac{t}{\tau_2}} \left(\frac{G}{\tau_2 \tau_1} \frac{\tau_2 \tau_1}{\tau_1 - \tau_2} e^{\frac{t(\tau_1 - \tau_2)}{\tau_2 \tau_1}} + C \right)$$

$$= e^{-\frac{t}{\tau_2}} \left(\frac{G}{\tau_2 \tau_1} \int e^{\frac{t(\tau_1 - \tau_2)}{\tau_2 \tau_1}} dt + C \right)$$

$$= e^{-\frac{t}{\tau_2}} \left(\frac{G}{\tau_1 - \tau_2} e^{\frac{t(\tau_1 - \tau_2)}{\tau_2 \tau_1}} + C \right)$$

$$\frac{t(\tau_1 - \tau_2)}{\tau_2 \tau_1} - \frac{t}{\tau_2 \tau_1} = -\frac{t(\tau_2)}{\tau_2 \tau_1} = -\frac{t}{\tau_1}$$

$$v_{c2} = \frac{G}{\tau_1 - \tau_2} e^{-\frac{t}{\tau_1}} + C e^{-\frac{t}{\tau_2}}$$

$$v_{c2}(0) = 0 \quad 0 = \frac{G}{\tau_1 - \tau_2} + C \quad C = -\frac{G}{\tau_1 - \tau_2}$$

$$v_{c2}(t) = \frac{G}{\tau_1 - \tau_2} \left(e^{-\frac{t}{\tau_1}} - e^{-\frac{t}{\tau_2}} \right), \quad t \geq 0$$

$$4) \quad s(t) = \int_0^t \frac{G}{\tau_1 - \tau_2} \left(e^{-s/\tau_1} - e^{-s/\tau_2} \right) ds$$

$$= \frac{G}{\tau_1 - \tau_2} \left[-\tau_1 e^{-s/\tau_1} + \tau_2 e^{-s/\tau_2} \right]_{s=0}^{s=t}$$

$$= \frac{G}{\tau_1 - \tau_2} \left[-\tau_1 e^{-t/\tau_1} + \tau_2 e^{-t/\tau_2} - (-\tau_1 + \tau_2) \right]$$

$$s(t) = \frac{G}{\tau_1 - \tau_2} \left[\tau_1 (1 - e^{-t/\tau_1}) + \tau_2 (e^{-t/\tau_2} - 1) \right]$$

$$5) \lim_{t \rightarrow \infty} s(t) = \frac{G}{\tau_1 - \tau_2} [\tau_1 - \tau_2] = G = 1 + \frac{R_5}{R_4}$$

$$6) \quad v_{in}(t) = A(u(t) - u(t-T)) = A u(t) - A u(t-T)$$

$$= A, \quad 0 \leq t < T$$

$$0, \quad T \leq t$$

$s(t)$ = response to unit step input

$$= \frac{G}{\tau_1 - \tau_2} [\tau_1 (1 - e^{-t/\tau_1}) + \tau_2 (e^{-t/\tau_2} - 1)]$$

$$A u(t) - A u(t-T) \rightarrow \boxed{} \rightarrow A s(t) - A s(t-T)$$

$$v_{out} = A [s(t) - s(t-T)]$$

prove LTI?

$$7) \quad s(t) = \frac{G}{\tau_1 - \tau_2} [\tau_1 (1 - e^{-t/\tau_1}) + \tau_2 (e^{-t/\tau_2} - 1)]$$

$$v_{out} = A [s(t) - s(t-T)]$$

$$= \frac{A G}{\tau_1 - \tau_2} \left([\tau_1 (1 - e^{-t/\tau_1}) + \tau_2 (e^{-t/\tau_2} - 1)] - [\tau_1 (1 - e^{-\frac{(t-T)}{\tau_1}}) + \tau_2 (e^{-\frac{(t-T)}{\tau_2}} - 1)] \right)$$

$$= \frac{A G}{\tau_1 - \tau_2} \left[\tau_1 \left(-e^{-\frac{t}{\tau_1}} + e^{-\frac{(t-T)}{\tau_1}} \right) + \tau_2 \left(e^{-\frac{t}{\tau_2}} - e^{-\frac{(t-T)}{\tau_2}} \right) \right]$$

$$8) \quad T \approx 0.02 \tau \quad (\text{for 1\% error}),$$

where τ is the smallest time constant

$$\text{smallest time constant: } \tau_2 = 4.7 \cdot 10^{-3} \text{ s}$$

$$T \approx 0.02 (4.7 \cdot 10^{-3} \text{ s}) = 9.4 \cdot 10^{-5} \text{ s}$$